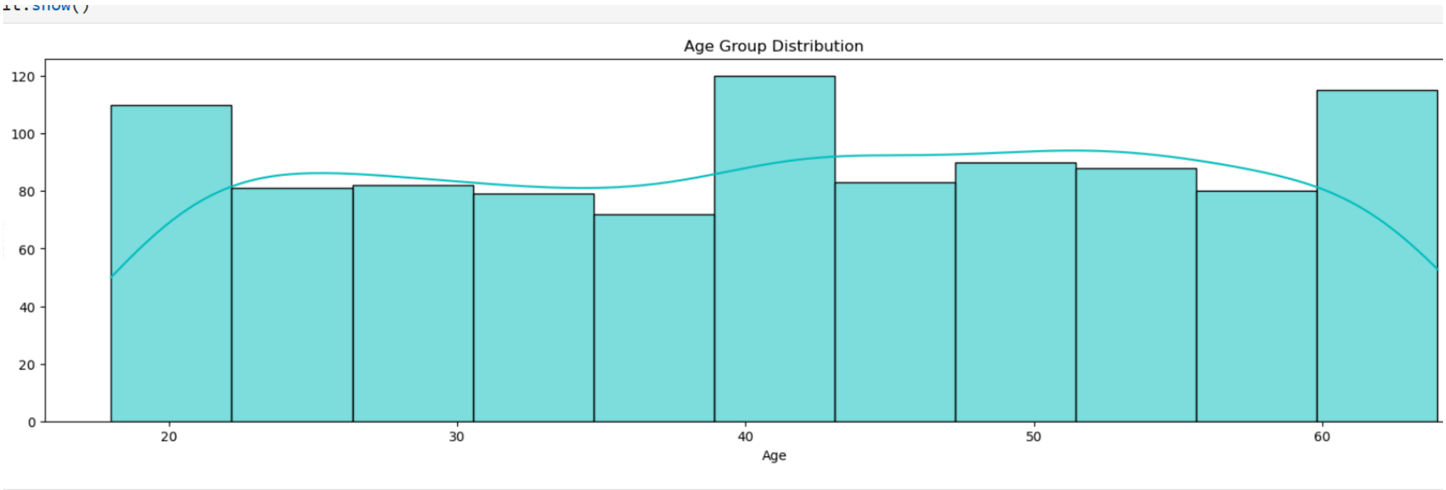
RETAIL SALES DATA ANALYSIS

01. Age Group Distribution Analysis



Observation:

- Customer ages range from approximately 18 to 64 years, indicating a broad customer base.
- The distribution is relatively uniform, with customers spread across most age ranges.
- Slightly higher customer concentrations are visible around the 40–55 age range.
- No single age group dominates the dataset, suggesting the business serves customers from multiple generations.
- The smooth KDE curve indicates there are no extreme age clusters or unusual spikes in customer age distribution.

Business Insight:

- Since customers are distributed across a wide range of ages, the business should avoid focusing on a single age segment.
- Marketing campaigns can be customized for different age groups (young adults, middle-aged customers, and older customers).
- The strong presence of customers between 40–55 years suggests this segment may represent a valuable target audience for premium products and loyalty programs.

02.GENDER DISTRIBUTION

OBSERVATION

- Female customers account for approximately 51% of total customers.
- Male customers account for approximately 49% of total customers.
- The customer base is nearly balanced across genders.

Business Insight:

- The retail business attracts both male and female customers equally, indicating broad market appeal.
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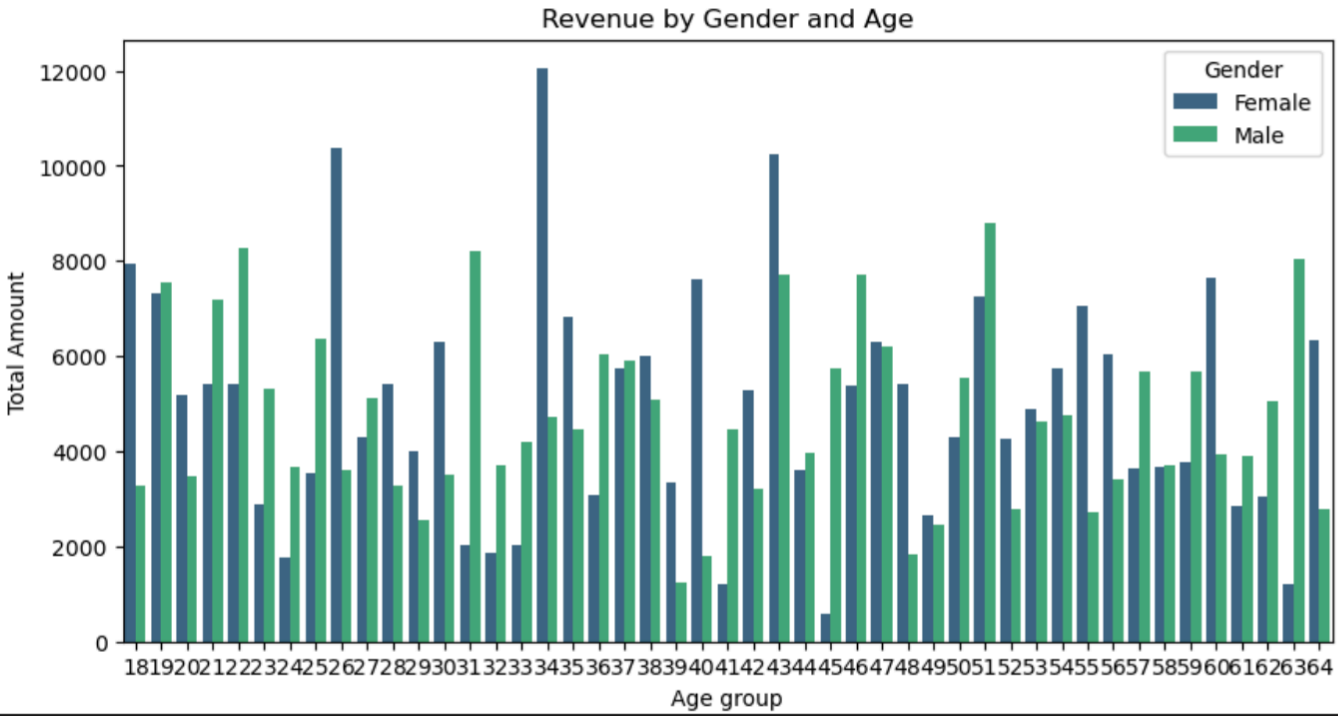
03. REVENUE BY AGE AND GENDER



Observation:

- Revenue varies considerably across ages and genders.
- Female customers generate higher revenue at some ages (e.g., around 26, 34, 43).
- Male customers generate higher revenue at some ages (e.g., around 22, 31, 51, 63).
- There is no consistent pattern showing one gender always spends more than the other.
- Spending behavior differs across demographic segments.

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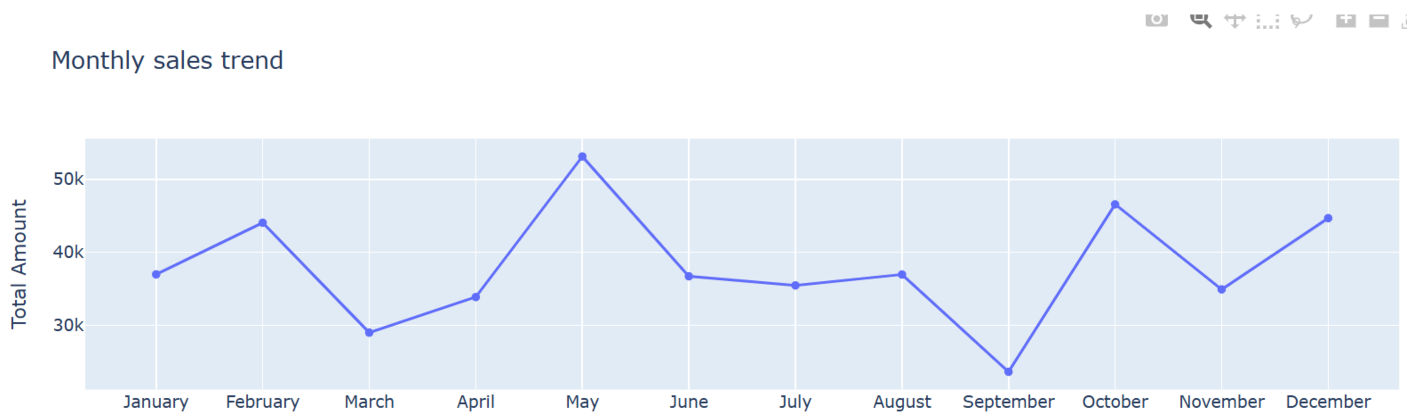
04. Monthly Sales Trend Analysis



Observation

- Sales fluctuate throughout the year, indicating seasonal variations in customer purchasing behavior.
- May recorded the highest sales, generating approximately 53K in revenue.
- September recorded the lowest sales, generating approximately 24K in revenue.
- Strong sales performance is also observed in October and December.
- Revenue declines after peak periods but recovers toward the end of the year.

Business Insight

- The company should investigate factors that contributed to strong sales in May, October, and December and replicate these strategies.
- Additional promotional campaigns should be introduced during low-performing months such as September.
- Inventory planning should account for seasonal demand fluctuations.

Key Finding

May was the best-performing month, while September experienced the lowest revenue, indicating clear seasonality in sales performance.

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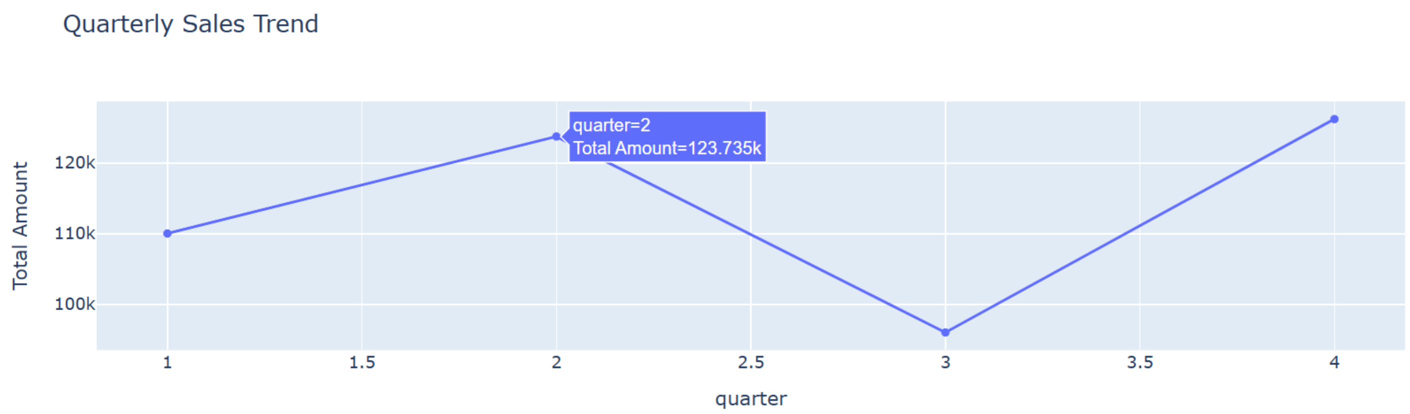
05. Quarterly Sales Trend Analysis



Observation

- Q4 generated the highest revenue, approximately 126K.
- Q2 was the second-best performing quarter with approximately 124K in sales.
- Q3 generated the lowest revenue, approximately 96K.
- Sales increased from Q1 to Q2, declined significantly in Q3, and recovered strongly in Q4.

Business Insight

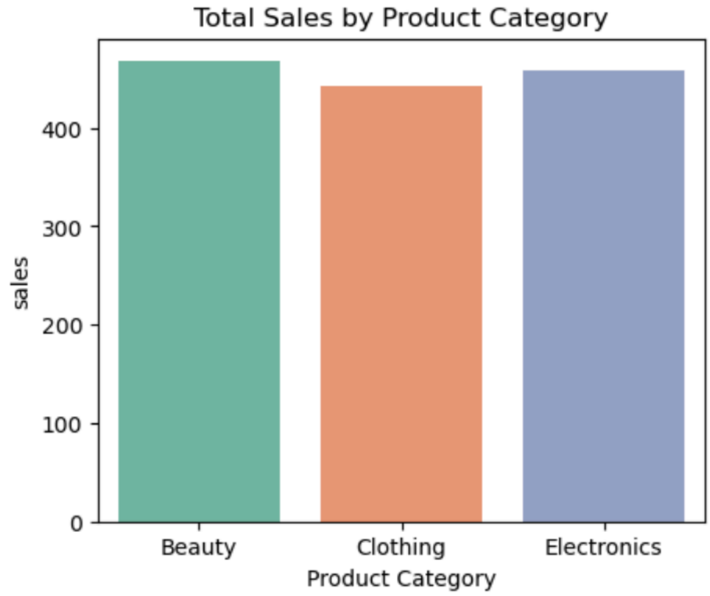
- Q4 appears to be the most profitable period and should receive increased marketing and inventory investment.
- The decline in Q3 suggests weaker customer demand or ineffective promotions during that period.

- Management should analyze Q3 performance and introduce targeted campaigns to improve sales.

Key Finding

Q4 was the highest-revenue quarter, while Q3 was the weakest. Strengthening sales strategies during Q3 could significantly improve annual revenue.

06. PRODUCT CATEGORY ANALYSIS BY REVEUE GENERATED



Observation

- Beauty products generate most revenue.
- Electronics also shows strong customer demand.
- Clothing has slightly fewer sales compared to the other categories.
- No category overwhelmingly dominates the market.

Business Insight

- Since all categories perform similarly, the business has a diversified revenue stream and is not dependent on a single product category.
- Beauty products could receive additional promotional focus since they currently attract the highest customer demand.
- Clothing may benefit from targeted discounts, bundle offers, or marketing campaigns to increase sales volume.

07.PRODUCT CATEGORY ANALYSIS BY QUANTITY

	Product Category	Quantity
1	Clothing	894
2	Electronics	849
0	Beauty	771

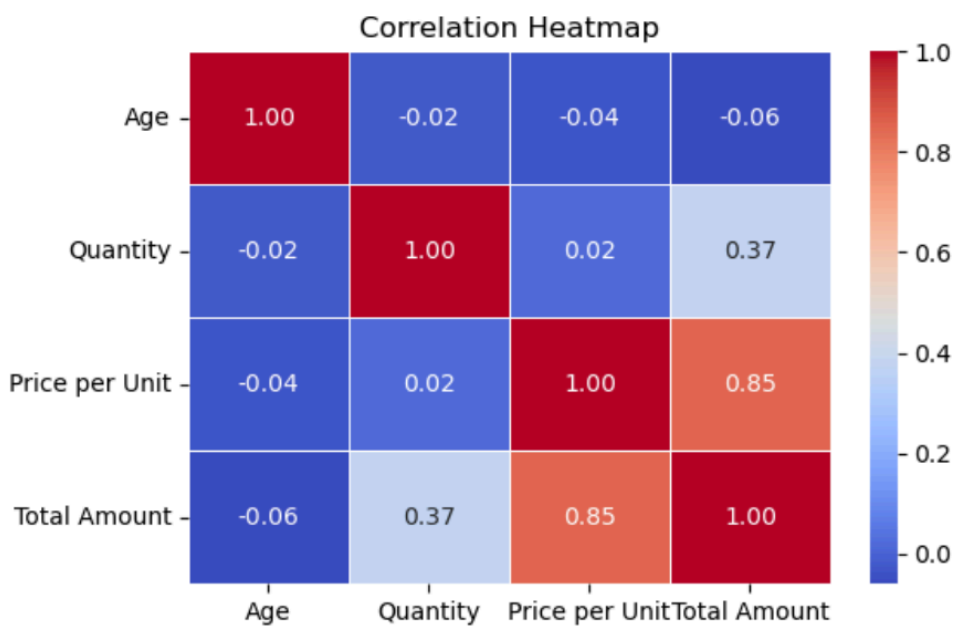
Key Finding

Clothing products are the most purchased category in terms of volume, while Beauty products have the lowest sales volume.

Business Insight

- High Clothing sales suggest customers buy apparel more frequently, making it a key category for maintaining customer engagement.
- Despite selling fewer units, Beauty products may still generate higher revenue if they have a higher average selling price.
- The business should ensure adequate inventory for Clothing products to avoid stock shortages.
- Cross-selling and promotional campaigns can be used to increase sales of Beauty products.

08.CORRELATION



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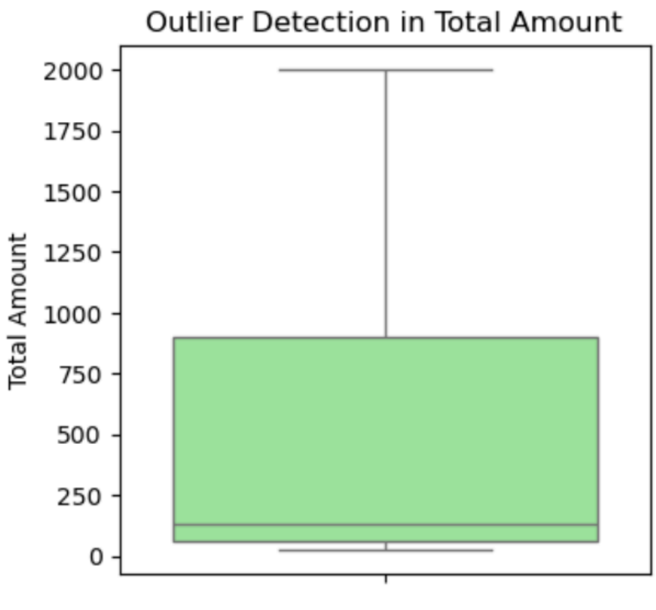
Key Findings:

- **Strongest Relationship (0.85): There is a very strong positive correlation between Price per Unit and Total Amount. This means that higher-priced premium items are the primary driver behind larger transaction totals.**
- **Moderate Relationship (0.37): There is a mild positive correlation between Quantity and Total Amount. Buying more items increases the bill, but not nearly as much as buying an expensive item does.**
- **No Relationship with Age (-0.02 to -0.06): Customer Age has near-zero correlation with any other metric (Quantity, Price per Unit, or Total Amount). An individual's age does not influence how many items they buy or how much they spend per transaction.**

Actionable Business Recommendations:

- **Prioritize High-Ticket Items: Because overall revenue (Total Amount) is heavily tied to item price rather than order volume, focus your marketing on upselling premium, higher-priced products to maximize transaction values.**
- **Rethink Age-Based Segmentation: Do not waste advertising budget creating different product price tracks or quantity bundles specifically based on a customer's age. Since shopping habits are uniform across all generations, broad behavioral promotions (like "Buy More, Save More") will work better than age-restricted ones.**

09.Outlier Detection: Total Amount



A boxplot was used to analyze the distribution of transaction amounts. No significant outliers were observed. The median transaction value is considerably lower than the maximum transaction value, indicating a right-skewed distribution where most purchases are of lower value while a few transactions contribute substantially higher revenue.

10.Outlier Detection: Price per Unit

There are no isolated dots outside the main lines (whiskers). The price distribution is statistically clean and free of abnormal pricing anomalies or entry errors.The middle line (median) sits very low, right around 50k. This tells us that the majority of individual item sales come from cheaper products, even though inventory prices scale all the way up to 500k.